EXECUTIVE SUMMARY

What The Backtest Is Actually Saying

This is a short decision memo, not the full report. It separates the investable core from the high-reward but hard-to-live-through ideas. The goal is simple: earn enough, survive the bad years, and avoid fooling ourselves with forward-looking bias.

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Current regime: Fragile Late-Cycle Watch

Quadrant: Growth Down / Inflation Down

History tested: 2006-2026

BOTTOM LINE

Keep The Unlevered Reserve Rule As Core

> The best balanced engine remains the unlevered quality rotation, but the best capital-preservation engine is the unlevered reserve rule. The x3 variants produce eye-catching terminal wealth and brutal drawdowns. A 90% drawdown is possible, and in the x3 SPY benchmark it effectively happened.

- Best unlevered growth engine: Quality Rotation, about +11.2% CAGR with -35.6% max drawdown.
- Best steadier-risk profile: Reserve Rule, about +10.0% CAGR with -31.5% max drawdown.
- Why the 2000% number appears: Quality x3 compounded from a much riskier path; it is a leveraged total-return figure, not an annual rate.
- What not to ignore: SPY x3 still ended near +5.4% CAGR but with -93.4% max drawdown.

PRINCIPLE 1

Protect The Downside First

The reserve rule earns less than the fastest strategy, but it keeps the path smoother. That matters because strategies only

PRINCIPLE 2

Borrow Only After You Earn The Right

Leverage is not alpha. It amplifies whatever quality already exists. In this dataset, the x3 variants raise reward and slash survivability.

PRINCIPLE 3

Separate Prediction From Proof

The model signal is one thing. The proof is the untouched holdout and the walk-forward history. They are both shown here, separately.

work if you can hold them through the bad times.

LOWER RISK / STEADIER REWARD

Reserve Rule

This is the capital-preservation bucket: smoother path, smaller drawdowns, slower wealth growth.

History CAGR +10.0% | Max DD -31.5% | Sharpe 0.84

BALANCED RISK / STRONG REWARD

Quality Rotation

This is the efficient middle: meaningful compounding without borrowing and without catastrophic path risk.

History CAGR +11.2% | Max DD -35.6% | Sharpe 0.82

HIGH RISK / HIGH REWARD

Reserve x3

Reward can be large, but the path is violent and hard to hold in live capital.

History CAGR +10.3% | Max DD -49.9% | Sharpe 0.58

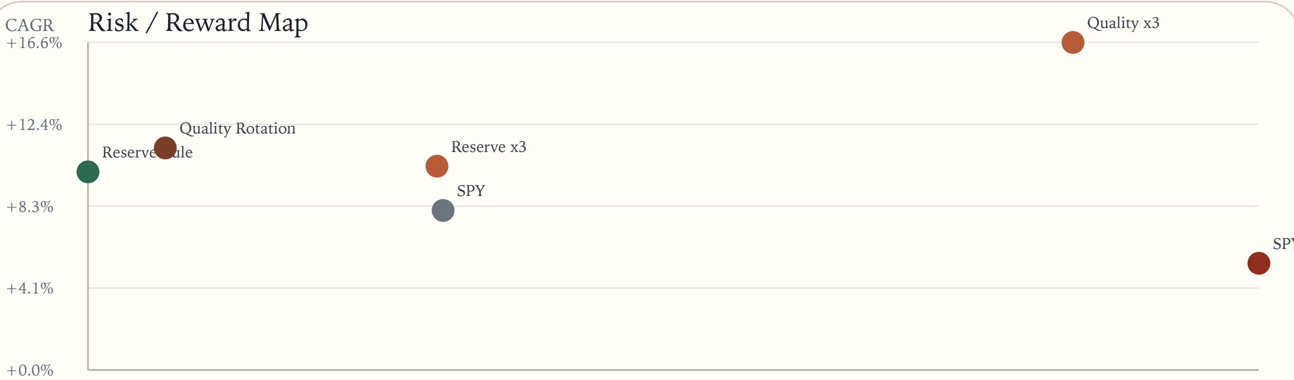
EXTREME RISK / WEAK REWARD QUALITY

SPY x3

Borrowing magnifies the pain faster than it improves the compounding path.

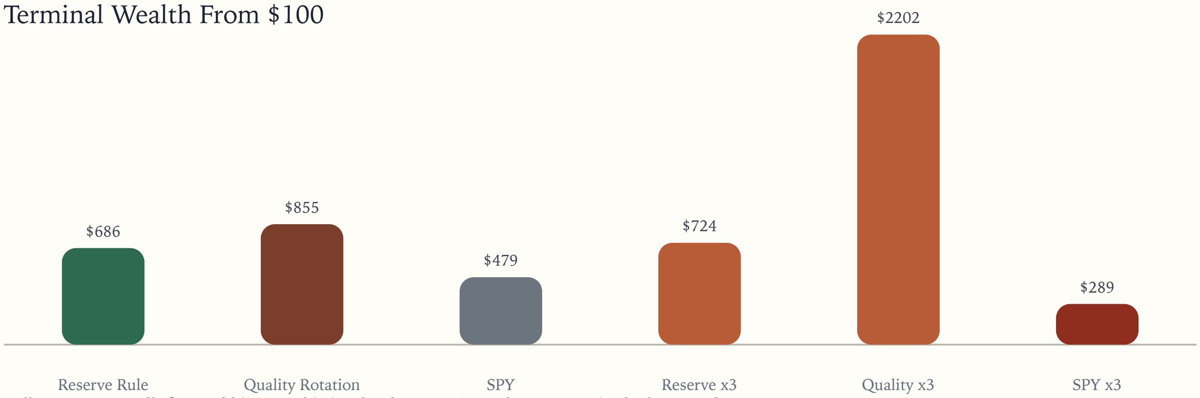
History CAGR +5.4% | Max DD -93.4% | Sharpe 0.36

Risk / Reward Map



Left is better: smaller drawdown. Up is better: higher CAGR.

Terminal Wealth From $100



Full 2006-2026 walk-forward history. This is why the 2000% numbers appear in the leveraged ML rows.

PAGE TWO

Simple Scorecard And Bias Controls

This page answers two questions. First: which ideas belong in the low-risk, balanced, and high-risk buckets? Second: why this is not a forward-looking backtest.

RISK LADDER

High Reward Is Not The Same As High Quality

STRATEGY	BUCKET	HISTORY CAGR	HISTORY MAX DD	SHARPE	HOLDOUT CAGR	HOLDOUT MAX DD
Reserve Rule	Lower risk / steadier reward	+10.0%	-31.5%	0.84	+8.3%	-1.6%
Quality Rotation	Balanced risk / strong reward	+11.2%	-35.6%	0.82	+11.0%	-3.0%
SPY	Medium risk / lower efficiency	+8.1%	-50.3%	0.58	+11.1%	-2.8%
Reserve x3	High risk / high reward	+10.3%	-49.9%	0.58	+8.6%	-1.6%
Quality x3	High risk / high reward	+16.6%	-83.6%	0.58	+25.1%	-10.4%
SPY x3	Extreme risk / weak reward quality	+5.4%	-93.4%	0.36	+26.1%	-10.2%

SIMPLE READ

How To Use The Buckets

- **Lower risk / steadier reward:** the reserve rule. This is the most usable sleeve for real capital because drawdown is controlled and turnover is low.
- **Balanced risk / strong reward:** the unlevered quality rotation. This is the best pure alpha engine, but it asks for more turnover and a rougher path.

- **High risk / high reward:** the reserve x3 sleeve and the quality x3 model. These are tactical ideas, not default policy portfolios.
- **Extreme risk / weak reward quality:** SPY x3. It proves that leverage alone is not a strategy.

> A 90% drawdown matters even if the final terminal wealth looks large. The path can disqualify the strategy even when the endpoint looks impressive.

Why This Is Not Forward-Looking

Train
5-year expanding window
features lagged 1 bar

Validate
1-year walk-forward test
purge 5 bars, embargo 5 bars

Holdout And Execution
untouched holdout starts 2025-01-01
trade one bar later, hold 5 bars

Macro regime labels are used to explain results, not to generate historical signals. Sector quality priors come only from the pre-holdout validation window. Reserve deployment uses only the SPY drawdown visible on the signal date. There is no future return, future regime, or future allocation leak in the benchmark path.

WHY THIS IS NOT FORWARD-LOOKING

Controls Against Bias

- Features are lagged one bar before training and prediction.
- Each walk-forward fold uses a 5-bar purge and a 5-bar embargo.
- The final holdout starts on 2025-01-01 and is never used for model selection.
- Signals are executed one bar after the signal date and held for 5 bars.
- Validation quality priors are estimated only from the pre-holdout window.
- Macro regime labels explain the results but do not generate the historical trades.

DECISION

What To Stick To

- **Core portfolio:** unlevered reserve cash rule.
- **Growth sleeve:** unlevered quality rotation if you can tolerate higher turnover and deeper drawdowns.
- **Tactical only:** reserve x3 drawdown sleeve, and only if you explicitly accept much deeper drawdowns.
- **Avoid as core:** SPY x3 and the full x3 rotation variants. They generate striking terminal wealth and unacceptable path risk.

> The safest honest reading is: there is evidence of edge, there is no evidence that leverage improves the quality of that edge enough to deserve core capital.